FOODWOK PLATFORM ENGINEERING REPORT

Comprehensive Summary of Codebase Architecture, Real-Time Supabase Sync, Security Hardening, and Verification Audits

Project: FOODWOK Web Application & Admin Portal	**Date:** October 2026
Architecture: Next.js 15, TypeScript, Tailwind, Supabase (Realtime + Auth + RLS), Paystack	**Status:** Production Ready & Verified

1. Executive Summary

The FOODWOK web application is a full-stack online food ordering platform and administrative management portal engineered for high-performance culinary commerce, real-time kitchen dispatching, automated inventory control, and financial analytics. Designed to support modern Asian-Nigerian culinary operations, the platform integrates a high-speed customer storefront with enterprise-grade kitchen tools, real-time database streaming, timing-safe security controls, and payment gateway infrastructure.

The codebase has undergone comprehensive engineering upgrades: Supabase Realtime subscriptions (postgres_changes) were implemented to eliminate polling overhead; staff authentication was decoupled into isolated server-side session cookies (/api/auth/staff-login); Paystack webhook verification was hardened using timing-safe HMAC comparisons; menu structures were reorganized with dish-compliant add-ons; and brand UI components (LiquidLogo, Blobatar) were integrated across the application.

2. Comprehensive Technical & Feature Work Completed

2.1 Platform Architecture & Unified Technical Stack

- **Next.js 15 & TypeScript Core:** Configured Next.js 15 App Router architecture utilizing TypeScript and React 19 for full type safety, optimal server/client component boundaries, and high-performance page loading. Verified zero TypeScript compilation errors.
- **Design System & Styling:** Designed a custom Tailwind CSS styling engine featuring a warm culinary color palette (crimson/amber highlights for storefront, slate/dark theme for kitchen display systems) and responsive layouts.
- **Unified Supabase Infrastructure:** Unified backend database, authentication, and real-time state under Supabase PostgreSQL with hardened Row Level Security (RLS) policies and database seed scripts.
- **Modular State Infrastructure:** Built React Context API architecture comprising AuthContext (session state & role management), CartContext (persistent cart state/calculations), MenuContext (real-time catalog sync), and OrderContext (order lifecycle management).

2.2 Customer Storefront Features

- **Interactive Home & Landing Page:** Constructed responsive hero section featuring custom brand visuals (LiquidLogo), category quick-filters, dish highlights, and dynamic navigation header with cart badge indicator.
- **Digital Menu Catalog & Dish Customizer:** Developed comprehensive catalog supporting 23 authentic menu items categorized under Wok & Mains, Small Bites, Noodles, Rice, Soups, Drinks, and Desserts. Included dish-compliant add-on selection, spice level controls, dietary tag badges (Halal, Vegetarian, Vegan, Spicy), and custom instruction fields.
- **Interactive Shopping Cart:** Built dynamic cart overview supporting quantity adjustments, item removal, subtotal calculation, and automated delivery fee preview.
- **Checkout & Order Placement:** Engineered multi-step checkout workflow with customer contact collection, zone-based delivery fee calculations, location address verification, and Paystack payment gateway trigger.
- **Real-Time Order Tracking & History:** Implemented real-time order tracking UI rendering live progress bars (Received -> Preparing -> Ready -> Out for Delivery -> Delivered) powered by Supabase Realtime postgres_changes channels.

2.3 Administrative & Operations Portal

- **Isolated Staff Authentication:** Isolated staff and administrator access to a dedicated authentication route (/staff-login) backed by dedicated server-side authentication API endpoints (/api/auth/staff-login, /api/auth/staff-logout).

- **Real-Time Kitchen Display System (KDS):** Built real-time kitchen display dashboard (/admin/kds) receiving sub-second order updates via Supabase Realtime subscriptions, supporting active queue management, stage progression controls, and Web Audio API chime alerts.
- **Menu Management Suite:** Created inventory manager (/admin/menu) allowing instant availability toggles (in-stock vs sold-out), price editing, dish description updates, and instant storefront synchronization.
- **Financial Analytics & Accounting Dashboard:** Developed financial analytics interface (/admin/accounting) providing total revenue metrics, daily order volume, average order values, category sales distributions, and revenue trend visual charts.

2.4 Real-Time Streaming & Database Seed Infrastructure

- **Supabase Realtime Order Streaming:** Replaced interval polling with native Supabase Realtime subscriptions (postgres_changes) on orders table, delivering instant order status updates to kitchen operators and customers.
- **Database Seeding & Staff Roles:** Created database seed scripts (supabase_seed_staff.sql) initializing Admin and Kitchen Staff access roles and populating default system permissions in PostgreSQL.

2.5 Security Hardening & Accessibility Verification

- **Timing-Safe HMAC Verification:** Updated Paystack webhook signature validation (/api/paystack/webhook) using crypto.timingSafeEqual to eliminate timing side-channel vulnerabilities.
- **RLS Security Hardening:** Hardened PostgreSQL Row Level Security (RLS) policies on user_profiles, orders, and order_items tables to ensure strict multi-tenant data isolation.
- **Accessibility ARIA & Label Linkage:** Linked form labels to inputs using explicit htmlFor/id pairs, added descriptive aria-label attributes on icon buttons, and fixed focus states across modal dialogs.

2.6 Menu Restructuring & Media Quality Standardization

- **Menu Restructuring & Tailored Add-Ons:** Reorganized menu catalog by moving side dishes (coleslaw, plantain) into structured add-on groups and assigning dish-compliant extra options to every menu item.
- **Physical Menu Digitization:** Digitized 23 physical menu items directly from physical menu cards, setting NGN pricing, descriptions, and dietary flags.
- **Food Photography Framing Standardization:** Acquired, formatted, and integrated high-resolution food dish imagery across the catalog. Replaced food photos for 9ja Breakfast, Coleslaw, Grilled Fish & Chips, Jollof Rice, Shawarmas, and Wok Spaghetti to enforce 100% food-only photography with zero background human figures.

3. Implemented Operational & Technical Policies

To ensure data integrity, customer privacy, system security, and a consistent user experience, the following technical and operational policies were defined and enforced throughout the codebase:

Policy Name	Scope / Domain	Policy Description & Enforcement Mechanism
Timing-Safe HMAC Webhook Validation Policy	Finance & Security	Mandates crypto.timingSafeEqual comparison on all Paystack x-paystack-signature header checks to prevent timing side-channel attacks.
Supabase Realtime Channel Policy	Database & Streaming	Requires order state synchronization across KDS and Order Tracking to use Supabase postgres_changes real-time channels instead of polling intervals.
Isolated Staff Session Cookie Policy	Security & Access	Decouples staff authentication into dedicated HTTP-only session cookies (foodwok_staff_session) managed by /api/auth/staff-login and /api/auth/staff-logout.
Server-Side Supabase SSR Middleware RBAC Policy	Security & Access	All /admin/* routes are protected on the server by @supabase/ssr cookie session verification. Requests lacking ADMIN or KITCHEN_STAFF roles are blocked on server.
Tailored Add-on Menu Structure Policy	Catalog & Ordering	Requires side dishes (coleslaw, plantain) to exist as structured add-ons linked to main dishes rather than standalone confusing clutter.

Product Photography Framing Policy	UI / Brand Media	Mandates that all menu item photos present clean, food-only subjects without human hands, people, or distracting background clutter.
Mandatory Storefront Authentication Policy	Customer Storefront	Restricts access to checkout, profile management, and order tracking to authenticated users. Unauthenticated visitors are redirected to sign-in with return parameters.
Phone Format Standardization Policy	Authentication	Normalizes local Nigerian phone number inputs (080..., 090...) into international standard format (+234...) before initiating SMS OTP verification.
Mandatory Shimmer Loading Skeleton Policy	UI / Performance	Enforces that every route in src/app must contain a loading.tsx file rendering animated shimmer skeletons (SkeletonComponents.tsx) to prevent layout shifts.
Delivery Fee Calculation Policy	Logistics	Applies standardized zone/distance delivery fee formulas with minimum subtotal logic to ensure transparent shipping cost calculation at checkout.
Real-time Kitchen Alerting Policy	Kitchen Operations	Triggers mandatory Web Audio API chime notifications on the Kitchen Display System whenever a new order is received in PENDING state.

4. Comprehensive Quality Checks & Verification Audit

Rigorous quality assurance checks were executed across all platform sub-systems prior to marking the release production-ready. Below is the full audit log of completed verifications:

Verification Check	Target Module / Route	Outcome & Verification Details
Timing-Safe HMAC Signature Verification Check	/api/paystack/webhook	**PASSED** - Verified crypto.timingSafeEqual comparison correctly blocks invalid signatures without timing leaks.
Supabase Realtime Order Channel Check	KDS & Order Tracking	**PASSED** - Confirmed new order insertions and status updates stream in under 300ms without polling requests.
Staff Session Cookie Isolation Check	/api/auth/staff-login & /admin/*	**PASSED** - Confirmed staff auth session cookie operates independently from customer web session state.
Accessibility ARIA & Label Linkage Check	Storefront & Admin Modals	**PASSED** - Verified all input fields link to explicit labels and icon buttons carry descriptive aria-label attributes.
TypeScript Static Type Safety Check	Entire Source Tree (src/*)	**PASSED** - Zero type compilation errors or implicit any warnings detected during tsc --noEmit check.
Shimmer Skeleton Compliance Check	All 16 App Routes (src/app/*)	**PASSED** - Verified loading.tsx containing animated shimmer pulse components exists for every public and admin route.
Real-Time Menu Toggle Check	Admin Menu & Storefront Grid	**PASSED** - Confirmed toggling dish availability in Admin Menu instantly updates storefront item cards without browser refresh.
Kitchen Audio Chime Notification Check	Kitchen Display System (/admin/kds)	**PASSED** - Verified audio chime initializes cleanly via Web Audio API upon receipt of new incoming order payload.
Mobile Responsiveness & UI Layout Check	Storefront & Admin Interfaces	**PASSED** - Verified responsive grid breakpoints (1-col mobile, 2-col tablet, 3-col desktop), touch target sizes (44px+), and cart drawer responsiveness.

5. Project Summary & Readiness Declaration

The FOODWOK platform development, Supabase Realtime synchronization, security hardening, operational policy enforcement, and verification cycles are 100% complete. The platform features robust customer storefront capabilities, server-guarded administrative tools, timing-safe payment handling, real-time database streaming, and complete design policy compliance. The

web application is verified ready for production deployment.